

4. Linear Regression and Randomized Experiments

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Jaewon (“Jay-one”) Yoo

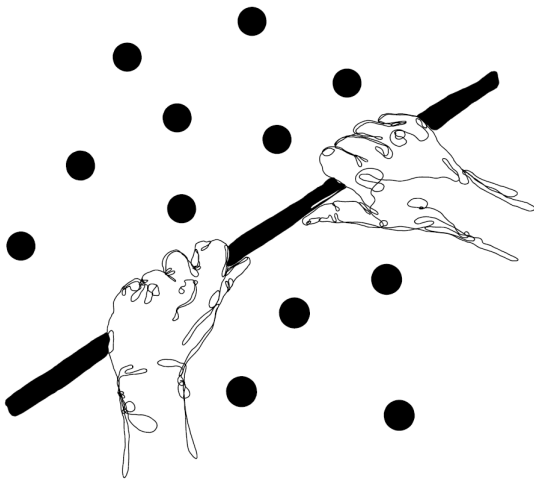
National Tsing Hua University

Roadmap

1. Regression with no covariates
2. Linear regression with covariates
3. Regression for stratified experiments
4. Cluster randomized experiments

Where are we? Where are we going?

- So far: analysis of experiments with Fisher's and Neyman's approaches.
 - Neyman: Unbiased estimators, (conservative) variances.
 - Fisher: exact test of the sharp null.
- Today: how does the workhorse estimator, OLS, fit into this story?
- Why would we consider using regression?
 - **Simplicity:** known tool that is already very common.
 - **Increased precision:** we may want to add covariates for more precise effect estimates.



Source: *Chapter 13 of The Effect (Textbook 2)* by Nick Huntington-Klein

1/ Regression with no covariates

Analysing Experiments with Regression?

- Q: Under complete randomization, can we use OLS to estimate ATEs
 - Literally, just $\text{lm}(y \sim d)$?
- Recall that the OLS estimator solves the least squares problem:

$$(\hat{\tau}_{\text{ols}}, \hat{\alpha}_{\text{ols}}) = \arg \min_{\tau, \alpha} \sum_{i=1}^n (Y_i - \alpha - \tau D_i)^2 \quad (1)$$

- The coefficient on a binary r.v. is mechanically the diff. in means:

$$\hat{\tau}_{\text{ols}} = \bar{Y}_1 - \bar{Y}_0 = \hat{\tau}_{\text{diff}} \quad (2)$$

- Standard Neyman analysis for unbiasedness, sampling variance.
- Generalized to discrete treatments with > 2 levels.

Justifying the Linear Model

- Mechanically the same, but can we justify the linear model itself?
 - Key assumptions: **linearity** and **mean independence of errors**.
- Some simple manipulation of the consistency assumption:

$$\begin{aligned}Y_i &= D_i Y_i(1) + (1 - D_i) Y_i(0) \\&= Y_i(0) + D_i \{Y_i(1) - Y_i(0)\} \\&= Y_i(0) + D_i \tau_i \\&= \mathbb{E}[Y_i(0)] + D_i \tau + \{Y_i(0) - \mathbb{E}[Y_i(0)]\} + D_i(\tau_i - \tau) \\&= \alpha + D_i \tau + \varepsilon_i\end{aligned}$$

- “Linear” functional form fully justified by consistency alone with:
 - Intercept $\alpha = \mathbb{E}[Y_i(0)]$ is the average control outcome.
 - Slope $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$ is the PATE.
 - Error is deviation for control PO + treatment effect heterogeneity.

Mean Independent Errors

$$\varepsilon_i = (Y_i(0) - \mathbb{E}[Y_i(0)]) + D_i \cdot (\tau_i - \tau)$$

- What about mean independence: $\mathbb{E}[\varepsilon_i|D_i] = 0$?
- Using randomization, we know D_i is independent of $Y_i(1)$, $Y_i(0)$, so:

$$\begin{aligned}\mathbb{E}[\varepsilon_i|D_i] &= \mathbb{E}[\{Y_i(0) - \mathbb{E}[Y_i(0)]\} + D_i \cdot (\tau_i - \tau)|D_i] \\&= \mathbb{E}[Y_i(0)|D_i] - \mathbb{E}[Y_i(0)] + D_i(\mathbb{E}[\tau_i|D_i] - \tau) \\&= \mathbb{E}[Y_i(0)] - \mathbb{E}[Y_i(0)] + D_i \underbrace{(\mathbb{E}[\tau_i] - \tau)}_{\tau = \mathbb{E}[\tau_i]} \\&= 0\end{aligned}$$

- Randomization + consistency $\rightsquigarrow$ linear model.
 - Does not imply homoskedasticity or normal errors, though!

Homoskedasticity

- Software default assumption: **Homoskedasticity**

$$\mathbb{V}[\varepsilon_i \mid \mathbf{D}] = \sigma^2, \quad \forall i$$

- But in general, based on previous error definition:

$$\mathbb{V}[\varepsilon_i \mid \mathbf{D}] = \mathbb{V}[\varepsilon_i \mid D_i] = D_i \sigma_1^2 + (1 - D_i) \sigma_0^2$$

- $\rightsquigarrow$ homoskedasticity true when $\sigma_1^2 = \mathbb{V}[Y_i(1)] = \mathbb{V}[Y_i(0)] = \sigma_0^2$
 - True under constant treatment effects!
- Under homoskedasticity, variance of the OLS estimator is:

$$\mathbb{V}[\hat{\tau}_{\text{ols}} \mid \mathbf{D}] = \frac{\sigma^2}{\sum_{i=1}^n (D_i - \bar{D})^2}$$

Variance Estimation

- “Standard” variance estimator under homoskedasticity:

$$\widehat{V}_{\text{const}} = \frac{\frac{1}{n-2} \sum_{i=1}^n \widehat{\varepsilon}_i^2}{\sum_{i=1}^n (D_i - \bar{D})^2} = \frac{\frac{1}{n-2} \sum_{i=1}^n (Y_i - \widehat{\alpha}_{\text{ols}} - \widehat{\tau}_{\text{ols}} D_i)^2}{\sum_{i=1}^n (D_i - \bar{D})^2}$$

- We can rewrite this as a function of the **pooled** variance $\widehat{\sigma}_{Y|D}^2$:

$$\widehat{V}_{\text{const}} = \widehat{\sigma}_{Y|D}^2 \left(\frac{1}{n_0} + \frac{1}{n_1} \right)$$

$$\widehat{\sigma}_{Y|D}^2 = \frac{1}{n-2} \left(\sum_{i:D_i=0} (Y_i - \bar{Y}_0)^2 + \sum_{i:D_i=1} (Y_i - \bar{Y}_1)^2 \right)$$

- **Inconsistent:** $\widehat{V}_{\text{const}} - \mathbb{V}[\widehat{\tau}] \xrightarrow{p} c \neq 0$ unless
 - Homoskedasticity holds: $\sigma_1^2 = \sigma_0^2$
 - Design is balanced: $n_1 = n_0$

Variance Estimation

- Can we use “standard” variance estimator: $\mathbb{V}[\varepsilon_i | \mathbf{D}] = \sigma^2$

- Inconsistent: $\widehat{\mathbb{V}}_{\text{const}} - \mathbb{V}[\widehat{\tau}] \xrightarrow{P} c \neq 0$

- Bias:

$$\begin{aligned} & \mathbb{E}(\widehat{\mathbb{V}}_{\text{const}}) - \mathbb{V}[\widehat{\tau}] \\ &= \underbrace{\mathbb{E}\left(\frac{\frac{1}{n-2} \sum_{i=1}^n \hat{\varepsilon}_i^2}{\sum_{i=1}^n (D_i - \bar{D})^2}\right)}_{\text{under const. effect assumption}} - \underbrace{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}\right)}_{\text{true variance}} \\ &= \frac{(n_1 - n_0)(n - 1)}{n_1 n_0 (n - 2)} (\sigma_1^2 - \sigma_0^2) \neq 0 \end{aligned}$$

- Unless:
 - Homoskedasticity holds: $\sigma_1^2 = \sigma_0^2$
 - Constant effect: $Y_i(1) - Y_i(0) = \text{const.}$
 - $\mathbb{V}[Y_i(1)] = \mathbb{V}[Y_i(0) + \text{const.}] = \mathbb{V}[Y_i(0)]$
 - Design is balanced: $n_1 = n_0$

Robust SEs

- Eicker-Huber-White (EHW) robust/sandwich variance estimator:

$$\underbrace{\widehat{\mathbb{V}}_{\text{EHW}}}_{\text{sandwich}} = \underbrace{\left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1}}_{\text{bread}} \underbrace{\left(\sum_{i=1}^n \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}_i' \right)}_{\text{meat}} \underbrace{\left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1}}_{\text{bread}}$$

$$= (\mathbb{X}'\mathbb{X})^{-1} \left(\sum_{i=1}^n \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}_i' \right) (\mathbb{X}'\mathbb{X})^{-1} \quad \text{where } \mathbb{X} = [\mathbf{1} \quad \mathbf{D}]$$

- Recall the PATE-targeted variance of the difference-in-means:

$$\mathbb{V}(\widehat{\tau}_{\text{diff}}) = \frac{\sigma_0^2}{n_0} + \frac{\sigma_1^2}{n_1} = \frac{\mathbb{V}[Y_i(0)]}{n_0} + \frac{\mathbb{V}[Y_i(1)]}{n_1}$$

- To see this, we can derive $\widehat{\mathbb{V}}_{\text{EHW}}$ under our case:

$$\widehat{\mathbb{V}}_{\text{EHW}} = \frac{\widetilde{\sigma}_1^2}{n_1} + \frac{\widetilde{\sigma}_0^2}{n_0}, \quad \text{where } \widetilde{\sigma}_d^2 = \frac{1}{n_d} \sum_{i:D_i=d} (Y_i - \bar{Y}_d)^2$$

- $\widetilde{\sigma}_0^2, \widetilde{\sigma}_1^2$ consistent for $\sigma_0^2, \sigma_1^2 \rightsquigarrow \widehat{\mathbb{V}}_{\text{EHW}}$ consistent for $\mathbb{V}(\widehat{\tau}_{\text{diff}})$

Better Robust SEs

- Many different “improved” versions of robust variances proposed.
 - Almost all are “finite-sample corrections” (no asymptotic effects)
- HC2 estimator normalizes residuals by the leverage, h_{ii} :

$$\mathbb{V}_{\text{HC2}} = (\mathbb{X}'\mathbb{X})^{-1} \left(\sum_{i=1}^n \frac{\hat{\varepsilon}_i^2}{1 - h_{ii}} \mathbf{x}_i \mathbf{x}_i' \right) (\mathbb{X}'\mathbb{X})^{-1}$$

- Leverage: $h_{ii} = \mathbf{x}_i (\mathbb{X}'\mathbb{X})^{-1} \mathbf{x}_i'$
- In this setting: $h_{ii} = n_1^{-1}$ if $D_i = 1$ and $h_{ii} = n_0^{-1}$ if $D_i = 0$
- Samii & Aronow (2012): HC2 is exactly the Neyman variance estimator:

$$\widehat{\mathbb{V}}_{\text{HC2}} = \frac{\widehat{\sigma}_0^2}{n_0} + \frac{\widehat{\sigma}_1^2}{n_1}$$

- $\rightsquigarrow$ simple OLS + HC2 = unbiased point and variance estimator.

Application in R Using (Sim) AI Exp. Data

```
1 > AI_data <- as_tibble(read.csv(url("https://bit.ly/3FHsusw"))); AI_data
2 # A tibble: 500 x 8
3   treat_ind test_outcome_pre test_outcome_post student_age student_gender tutor_age
4   <int>      <int>          <int>          <int>      <int>      <int>
5 1         1          0            1           11         0         27
6 2         1          0            1            6         1         45
7 3         0          1            1           15         1         54
8 4         0          1            1            6         1         50
9 5         0          1            0            8         1         64
10 6         1          1            1           15         1         43
11 7         1          0            1            7         1         47
12 8         1          1            1           16         0         49
13 9         0          0            1           12         1         42
14 10        1          1            1           11         1         28
15 # ... 490 more rows
16 # ... 2 more variables: years_of_experience <int>, education_level <chr>
17 # Use `print(n = ...)` to see more rows
18
19 > lm1 <- lm(test_outcome_post ~ treat_ind, data = AI_data)
20 > vcovM <- sandwich::vcovHC(lm1, type = 'HC2')
21 > sqrt(vcovM[1,1])
22 [1] 0.02955412
23 > sqrt(vcovM[2,2]) # sqrt(diag(vcovM))
24 [1] 0.04209018
25
26 > # Or
27 > estimatr::lm_robust(test_outcome_post ~ treat_ind, AI_data, se_type = 'HC2')
28       Estimate Std. Error  t value    Pr(>|t|)    CI Lower CI Upper  DF
29 (Intercept) 0.65134100 0.02955412 22.038927 1.233662e-75 0.59327487 0.7094071 498
30 treat_ind    0.03903557 0.04209018  0.927427 3.541541e-01 -0.04366065 0.1217318 498
```

2/ Linear regression with covariates

Adding Covariates

- What if we add covariates to our regression model?

$$(\hat{\tau}_{\text{adj}}, \hat{\alpha}_{\text{adj}}, \hat{\beta}_{\text{adj}}) = \arg \min_{\tau, \alpha, \beta} \sum_{i=1}^n (Y_i - \alpha - \tau D_i - \tilde{\mathbf{X}}_i' \beta)^2$$

- $\tilde{\mathbf{X}}_i = \mathbf{X}_i - \bar{\mathbf{X}}$ are **centered** covariates for notational ease.
- Why might we do this? To increase **precision** of our estimates.
 - We hope $\mathbb{V}[\hat{\tau}_{\text{adj}}] < \mathbb{V}[\hat{\tau}_{\text{diff}}]$ so we have smaller CIs, more powerful tests.
 - Intuition: less residual variation in Y_i after accounting for $\mathbf{X}_i$
- Questions:
 - Is $\hat{\tau}$ still unbiased? Consistent?
 - Should we expect an increase in precision?
 - Controversial! Freedman (2008) “Randomization does not justify the regression model”

OLS is biased, but consistent (Freedman, 2008. Adv. in Appl. Math)

- Agnostic approach: don't assume correctness of the linear model.
 - $\mathbf{X}_i'\beta$ could be badly misspecified.
 - No assumptions about homoskedasticity.
- Under minimal assumptions, OLS is consistent for the linear projection

$$(\widehat{\tau}_{\text{adj}}, \widehat{\alpha}_{\text{adj}}, \widehat{\beta}_{\text{adj}}) \xrightarrow{P} (\tau_0, \alpha_0, \beta_0) = \arg \min_{(\tau, \alpha, \beta)} \mathbb{E} [(Y_i - \alpha - \tau D_i - \widetilde{\mathbf{X}}_i'\beta)^2]$$

- $\widehat{\tau}_{\text{adj}}$ now **biased** for τ though bias should be small.
- But $\widehat{\tau}_{\text{adj}}$ is **consistent**.
 - Intuition: Since $D_i \perp\!\!\!\perp \mathbf{X}_i$, including $\widetilde{\mathbf{X}}_i$ won't (asymptotically) affect coefficient on D_i .
- Freedman (2008) shows the same thing for finite-sample inference.

Variance of Adjustment Estimator

- Complete randomization + single, mean-zero covariate X_i
 - Generalizes easily to more covariates.
 - Let $\sigma_{0x} = \text{cov}(Y_i(0), X_i)$ and $\sigma_{1x} = \text{cov}(Y_i(1), X_i)$.
 - Probability of treatment $p = n_1/n$
- Freedman (2008) derived gains from adjusting for X_i using OLS:

$$\mathbb{V}[\widehat{\tau}_{\text{diff}}] - \mathbb{V}[\widehat{\tau}_{\text{adj}}] = \frac{\sigma_{0x}\{\sigma_{0x} + 2(1 - 2p)\sigma_{1x}\}}{np(1 - p)}$$

- Will adjustment decrease the sampling variance?
 - If design is balanced, $p = 1/2$, then adjustment always helps.
 - Design imbalance could lead to adjustment hurting.
- Estimation: EHW robust variance estimators are consistent or asymptotically conservative for $\mathbb{V}[\widehat{\tau}_{\text{adj}}]$

Regression with Full Interactions

- OLS estimator from fully interacted model, $\widehat{\tau}_{\text{inter}}$:

$$Y_i = \alpha + \tau D_i + \widetilde{\mathbf{X}}_i' \beta + D_i \widetilde{\mathbf{X}}_i' \gamma + \varepsilon_i$$

- Equivalent to running separate Y_i on $\widetilde{\mathbf{X}}_i$ in each D_i
- As with non-interacted model, $\widehat{\tau}_{\text{inter}}$ is consistent for τ and asymptotically normal.
- Lin (2013): **fully interacted model will never hurt precision asymptotically.**
 - Freedman critique was right, but Lin shows an easy way to resolve.
- EHW robust variance estimators are consistent or asymptotically conservative.

Linear Regression with Covariates

In R

```
# Step 1: Compute centered covariates
your_data$Xtilde <- NULL

# Step 2: Write down your formula
your_formula <- NULL

# Step 3: Fit the model using lm() or estimatr::lm_robust()
your_fitted_model <- lm(your_formula, data = your_data)

# Step 4: Compute robust standard errors (skip if you used lm_robust)
your_vcov <- sandwich::vcovHC(your_fitted_model, type = 'HC2')

# Step 5: Check the point and se estimate of your coefficients
# (look for tau hat!)
est <- cbind("coef" = your_fitted_model$coef,
             "se" = sqrt(diag(your_vcov)))
```

Example Code

```
1 > AI_data <- AI_data |>
2   mutate(Xtilde = student_age - mean(student_age)) |>
3   select(treat_ind, test_outcome_post, Xtilde); head(AI_data,3)
4 # A tibble: 3 x 3
5   treat_ind test_outcome_post Xtilde
6     <int>         <int>    <dbl>
7 1         1             1 -0.566
8 2         1             1 -5.57
9 3         0             1  3.43
10
11 > estimatr::lm_robust(test_outcome_post ~ treat_ind * Xtilde, data = AI_data)
12               Estimate Std. Error   t value    Pr(>|t|)    CI Lower    CI Upper   DF
13 (Intercept)    0.649860304 0.029645673 21.9209157 5.557487e-75 0.591613722 0.708106886 496
14 treat_ind      0.038364943 0.042245496  0.9081428 3.642438e-01 -0.044637245 0.121367131 496
15 Xtilde         0.008351571 0.008581795  0.9731730 3.309417e-01 -0.008509581 0.025212723 496
16 treat_ind:Xtilde -0.019462905 0.012003266 -1.6214674 1.055529e-01 -0.043046422 0.004120612 496
17
18 > your_fitted_model <- lm(test_outcome_post ~ treat_ind * Xtilde, data = AI_data)
19 > vcovM_adj <- sandwich::vcovHC(your_fitted_model, type = 'HC2'); sqrt(diag(vcovM_adj))
20               (Intercept)      treat_ind      Xtilde treat_ind:Xtilde
21               0.029645673      0.042245496      0.008581795      0.012003266
22
23 > est <- cbind("coef" = your_fitted_model$coef,
24              "se" = sqrt(diag(your_vcov))); est
25               coef      se
26 (Intercept)    0.649860304 0.029645673
27 treat_ind      0.038364943 0.042245496
28 Xtilde         0.008351571 0.008581795
29 treat_ind:Xtilde -0.019462905 0.012003266
```

Summarizing Regression

- Regression with no covariates = standard Neyman analysis.
- Regression with (uninteracted) covariates:
 - Consistent for SATE/PATE.
 - Usually will help with precision, but can hurt.
- Regression with interacted covariates:
 - Consistent for SATE/PATE.
 - Asymptotically will never hurt precision.
- Always use robust/HC2 variance estimators unless you have good reasons.

3/ Regression for stratified experiments

Cf. Imbens & Rubin (2015), *Causal Inference for Statistics, Social, and Biomedical Sciences*, Ch. 9.6

Regression for Stratified Experiments

- Setup: block randomized experiment with block indicators W_{ij} .
 - Block “fixed effects” $W_{ij} = 1$ if i is in block j , 0 otherwise.
 - Blocks $j \in \{1, \dots, J\}$ with sizes $w_j = n_j/n$ and propensity scores $p_j = n_{1,j}/n_j$.
- STAR Project (Krueger, 1999): within each school (block), classes were randomized.
- Naive approach: just include the block FEs in OLS

$$(\hat{\tau}_{b,fe}, \hat{\alpha}_1, \dots, \hat{\alpha}_J) = \arg \min_{(\tau, \alpha_1, \dots, \alpha_J)} \sum_{i=1}^n \left(Y_i - \tau D_i - \sum_{j=1}^J \alpha_j W_{ij} \right)^2$$

- $\hat{\tau}_{b,fe}$ **not consistent** for the PATE unless:

$$\hat{\tau}_{b,fe} \xrightarrow{p} \frac{\sum_{j=1}^J \omega_j \tau_j}{\sum_{j=1}^J \omega_j}, \quad \text{where} \quad \omega_j = w_j p_j (1 - p_j)$$

- Propensity scores are equal across blocks: $p_j = p$ for all j .
- ATEs are equal across strata $\tau_j = \tau$ for all j .

Block Randomized Trials: Correct Analysis

1. Just use original Neyman analysis aggregating within-strata analyses.
 2. Weight OLS by inverse of the propensity score: $1/p_j$.
 3. Fully interact block FEs with treatment.
- Latter two allow for additional covariates to be added.
 - Check Imbens and Rubin (2015) Ch.9.6.1., second model
 - See this simulation study using DeclareDesign:
<https://declaredesign.org/blog/posts/biased-fixed-effects.html>

Block Randomized Trials: Correct Analysis

2. Weight OLS by inverse of the propensity score.

$$(\hat{\tau}_{b,fe}, \hat{\alpha}_1, \dots, \hat{\alpha}_J) = \arg \min_{(\tau, \alpha_1, \dots, \alpha_J)} \sum_{i=1}^n s_i \left(Y_i - \tau D_i - \sum_{j=1}^J \alpha_j W_{ij} \right)^2$$

$$\text{where } s_i = \sum_{j=1}^J \left\{ \left(\frac{1}{p_j} \right) D_i + \left(\frac{1}{1 - p_j} \right) (1 - D_i) \right\} W_{ij} \quad \text{and} \quad p_j = \frac{n_{1j}}{n_j}.$$

In R

```
your_formula <- as.formula("outcome ~ treat + x_tilde1 + x_tilde2")

your_data <- data.frame(outcome, treat,
                        x_tilde1, x_tilde2,
                        weights, block)

your_fitted_model <- estimatr::lm_robust(your_formula, data = your_data,
                                         weights = weights, # s
                                         se_type = "HC2",
                                         fixed_effects = block)
```

4/ Cluster randomized experiments

Cf. Angrist & Pischke (2008), *Mostly Harmless Econometrics*, Ch. 8

Clustering Treatments

- Treatment often allocated at a higher level than the data.
 - Counties are treated, but we have individual-level data.
 - Classrooms are treated, but we have student data.
- Has considerable benefits:
 - Often cheaper/easier to implement than individual assignment.
 - Allows for interference within clusters without bias.
- But lots of confusion about how to analyze.
 - More valuable to add more individuals or clusters?
 - What to do with individual-level covariates?

Cluster Randomized Trials

- Setup:
 - Clusters: $k \in \{1, \dots, K\}$
 - Randomly choose K_1 treatment clusters, K_0 control.
 - Each cluster has units $i \in \{1, \dots, m_k\}$ with $\sum_{k=1}^K m_k = n$
 - Treatment assignment at cluster level: $D_{ik} = D_k$
 - Potential outcomes $Y_{ik}(d)$
- Random assignment at the cluster level: $\{Y_{ik}(1), Y_{ik}(0)\} \perp\!\!\!\perp D_k$.
- Quantity of interest still at individual level:

$$\text{SATE} = \frac{1}{n} \sum_{k=1}^K \sum_{i=1}^{m_k} \{Y_{ik}(1) - Y_{ik}(0)\}$$

Analysis of Clustered Experiments

- Simple setting: all clusters have the same size $m_k = m$ for all k .
- Simple difference in means is unbiased:

$$\begin{aligned}\widehat{\tau}_{\text{cl}} &= \frac{1}{mK_1} \sum_{k=1}^K \sum_{i=1}^{m_k} D_k Y_{ik} - \frac{1}{mK_0} \sum_{k=1}^K \sum_{i=1}^{m_k} (1 - D_k) Y_{ik} \\ &= \frac{1}{K_1} \sum_{k=1}^K D_k \bar{Y}_k - \frac{1}{K_0} \sum_{k=1}^K (1 - D_k) \bar{Y}_k\end{aligned}$$

- $\bar{Y}_k$ is the cluster average: $\frac{1}{m} \sum_{i=1}^m Y_{ik}$
 - Unbiasedness follows from Neyman-style analysis at cluster level.
 - Estimator is biased, but consistent (in K) if cluster size varies.
- Neyman-style conservative variance:

$$\mathbb{V}[\widehat{\tau}_{\text{cl}} \mid \mathbf{D}] \leq \frac{\mathbb{V}[\bar{Y}_k(1)]}{K_1} + \frac{\mathbb{V}[\bar{Y}_k(0)]}{K_0} \quad \text{where } \bar{Y}_k(d) = \frac{1}{m} \sum_{i=1}^m Y_{ik}(d)$$

Cluster Robust Standard Errors

- What if we want to use OLS at individual level?
 - Adding individual and unit controls.
- **Cluster-robust variance estimator** for OLS:

$$\widehat{\mathbb{V}}_{\text{cl}}[\widehat{\alpha}, \widehat{\tau}] = (\mathbb{X}'\mathbb{X})^{-1} \left(\sum_{k=1}^K \mathbb{X}'_k \widehat{\boldsymbol{\varepsilon}}_k \widehat{\boldsymbol{\varepsilon}}'_k \mathbb{X}_k \right) (\mathbb{X}'\mathbb{X})^{-1}$$

- $\mathbb{X}' = [1 \quad \mathbf{D}]'$, $\mathbb{X}'_k = [1 \quad \mathbf{D}_k]'$, and $\widehat{\boldsymbol{\varepsilon}}_k = (\widehat{\varepsilon}_{1k}, \dots, \widehat{\varepsilon}_{m_k k})$
 - Consistent as the number of clusters grows.
- **Cluster at the treatment assignment level** (no higher or lower).
- Vanilla CRVE is biased, Bell & McCaffrey proposed CR2 adjustment similar to HC2 (usually preferable).

Cluster Randomized Trials: Analysis

In R

```
your_formula <- as.formula("outcome ~ treat + x_tilde1 + x_tilde2")

your_data <- data.frame(outcome, treat,
                        x_tilde1, x_tilde2,
                        cluster)

your_fitted_model <- estimatr::lm_robust(your_formula, data = your_data,
                                         clusters = cluster,
                                         se_type = "CR2")

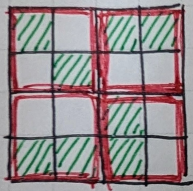
??estimatr::lm_robust # Check more options for se_type

# Or
your_model <- lm(your_formula, data = your_data)
your_vcov <- clubSandwich::vcovCR(your_model, cluster = your_data$cluster,
                                  type = "CR2")
```

- You may have block and cluster design at the same time.

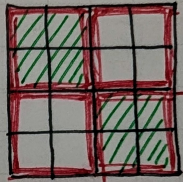
/// : treated

□ : block



/// : treated

□ : cluster



Onto the presentations & discussions!

Contact Information:

jaewon.yoo@iss.nthu.edu.tw

<https://j1yoo.github.io/>



Appendix

Cost of Clustering

- Standard variance under **individual assignment**:

$$\mathbb{V}[\hat{\tau}_{\text{diff}}] = \frac{\mathbb{V}[Y_{ik}(1)]}{mK_1} + \frac{\mathbb{V}[Y_{ik}(0)]}{mK_0}$$

- Under clustering, the variance of the cluster average inflates:

$$\frac{\mathbb{V}[\bar{Y}_k(1)]}{K_1} = \frac{\mathbb{V}[Y_{ik}(1)]}{mK_1} (1 + (m-1)\rho_1)$$

- Intraclass correlation coefficient** (ICC):

$$\rho_1 = \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{j \neq i} \mathbb{E} \left\{ (Y_{ik}(1) - \bar{Y}(1))(Y_{jk}(1) - \bar{Y}(1)) \right\}$$

- ρ measures how similar units are within clusters.
 - Usually cluster is less efficient because $\rho > 0$.
 - More similarity $\rightsquigarrow$ each unit provides redundant information $\rightsquigarrow$ less efficiency under clustering.

Linear Regression and Causality

- Regression: **conditional expectation function** of Y given $\mathbf{X}$

$$\mathbb{E}(Y|\mathbf{X}) = f(\mathbf{X}) = \beta^T \mathbf{X}$$

- Q: When can we interpret coefficients as causal effects?
- Causal model as structural equation model:

$$Y_i(d) = \alpha + \tau d + \varepsilon_i \quad \text{for } d = 0, 1, \text{ where } \mathbb{E}(\varepsilon_i) = 0$$

1. No interference between units
 2. $\mathbb{E}(Y_i(0)) = \alpha$
 3. $Y_i(1) - Y_i(0) = \tau$ for all $i \rightsquigarrow$ **constant unit-level causal effect**
- Heterogeneous treatment effect model:

$$Y_i(d) = \alpha + \tau_i d + \varepsilon_i = \alpha + \tau d + \underbrace{(\tau_i - \tau)d}_{=\varepsilon_i(d)} + \varepsilon_i$$

where $\mathbb{E}(\varepsilon_i) = 0$ and $\tau = \mathbb{E}(\tau_i) = \mathbb{E}(Y_i(1) - Y_i(0))$

- $\mathbb{E}(\varepsilon_i(d)) = 0$ for $d = 0, 1$
- $\alpha = \mathbb{E}(Y_i(0))$